UNIT II
Cryptographic Tools

Introduction

Symmetric encryption also referred to as conventional encryption or single-key encryption, as the only type of encryption in use prior to the development of public key encryption in the 1970s. It remains by far the most widely used of the two types of encryption.

Before beginning, we define some terms. An original message is known as the plaintext, while the coded message is called the cipher text. The process of converting from plaintext to cipher text is known as **enciphering or encryption**; restoring the plaintext from the cipher text is deciphering or decryption. The many schemes used for encryption constitute the area of study known as cryptography. Such a scheme is known as a cryptographic system or a cipher. Techniques used for deciphering a message without any knowledge of the enciphering details fall into the area of cryptanalysis. Cryptanalysis is what the layperson calls "breaking the code." The areas of cryptography and cryptanalysis together are called cryptology.

Symmetric Cipher Model

A symmetric encryption scheme has five ingredients as indicated in following figure

- **Plaintext:** This is the original intelligible message or data that is fed into the algorithm as input.
- **Encryption algorithm:** The encryption algorithm performs various substitutions and transformations on the plaintext.
- **Secret key:** The secret key is also input to the encryption algorithm. The key is a value independent of the plaintext and of the algorithm. The algorithm will produce a different output depending on the specific key being used at the time. The exact substitutions and transformations performed by the algorithm depend on the key.
- **Cipher text:** This is the non-readable/scrambled message produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different cipher texts. The cipher text is an apparently random stream of data and, as it stands, is unintelligible.

- **Decryption algorithm:** This is essentially the encryption algorithm run in reverse. It takes the cipher text and the secret key and produces the original plaintext.

  There are two requirements for secure use of conventional encryption:

  1. We need a strong encryption algorithm. At a minimum, we would like the algorithm to be such that an opponent who knows the algorithm and has access to one or more cipher texts would be unable to decipher the cipher text or figure out the key.
  2. Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure. If someone can discover the key and knows the algorithm, all communication using this key is readable.

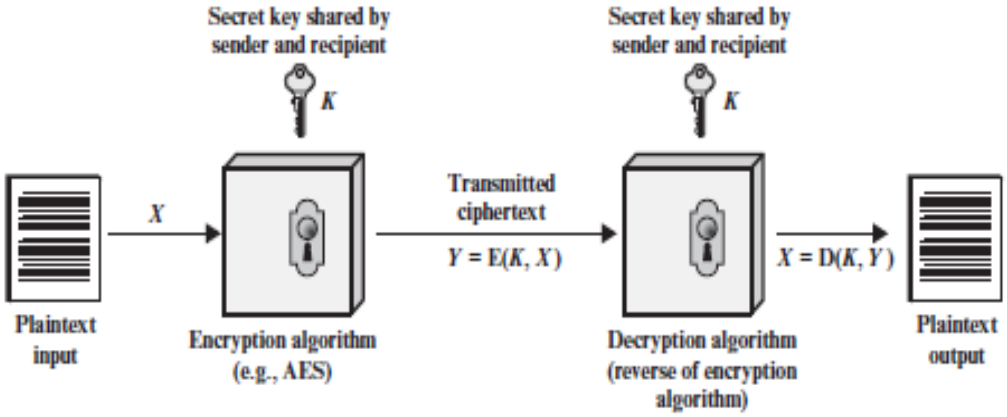

Fig.2.1: Simplified Model of Conventional Encryption

Let us take a closer look at the essential elements of a symmetric encryption scheme, using above figure 2.1 A source produces a message in plaintext, X = [X1, X2… XM]. The M elements of X are letters in some finite alphabet. For encryption, a key of the form K = [K1, K2… KJ] is generated. If the key is generated at the message source (sender), then it must also be provided to the destination (receiver) by means of some secure channel.

Alternatively, a third party could generate the key and securely deliver it to both source and destination. With the message X and the encryption key K as input, the encryption algorithm forms the cipher text Y = [Y1, Y2 … YN]. We can write this as Y = E (K, X)

This notation indicates that Y is produced by using encryption algorithm E as a function of the plaintext X, with the specific function determined by the value of the key K.

The intended receiver, in possession of the key, is able to invert the transformation: X = D (K, Y)

An opponent, observing Y but not having access to K or X, may attempt to Recover X or K or both X and K. It is assumed that the opponent knows the encryption (E) and decryption (D) algorithms. If the opponent is interested in only this particular message, then the focus of the effort is to recover X by generating a plaintext estimate X. Often, however, the opponent is interested in being able to read future messages as well, in which case an attempt is made to recover K by generating an estimate K.

Asymmetric Key Encryption/ Public Key Encryption

Asymmetric key encryption, also known as public-key encryption, is a type of encryption where two different keys are used: one for encryption (public key) and another for decryption (private key). This is in contrast to symmetric key encryption, where the same key is used for both encryption and decryption.

Key Features of Asymmetric Key Encryption:

1. **Public and Private Keys:**

- **Public Key:** This key is shared openly with anyone. It is used to encrypt data.
- **Private Key**: This key is kept secret and is used to decrypt data that was encrypted with the corresponding public key.

  The relationship between the two keys is mathematically linked, but it is computationally infeasible to derive the private key from the public key.

2. **Security:**
   - **Encryption:** A sender uses the recipient's public key to encrypt a message.
   - **Decryption:** The recipient uses their private key to decrypt the message.
   - Since the public key is shared widely and the private key is kept secure, only the recipient with the private key can decrypt the message.
3. **Digital Signatures:**
   - Asymmetric encryption can also be used for digital signatures. The sender can sign data with their private key, and anyone with the sender's public key can verify the authenticity and integrity of the data.
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4. **Key Pair:**
    - Asymmetric encryption relies on a pair of mathematically related keys (public and private). For example, in the RSA algorithm, if one key is used to encrypt the data, only the corresponding key can decrypt it.

How It Works (Simplified Process):

1. **Step 1:** The recipient generates a public-private key pair.
    - The public key is shared with others (it can be distributed openly).
    - The private key is kept secure and secret.
2. **Step 2:** The sender uses the recipient's public key to encrypt the message. Once encrypted, the message cannot be read by anyone except the recipient who possesses the corresponding private key.
3. **Step 3:** The recipient uses their private key to decrypt the encrypted message.

Advantages of Asymmetric Encryption:

- **Key Distribution:** No need to share a secret key in advance (as in symmetric encryption). Only the public key is exchanged, which can be shared freely.
- **Digital Signatures:** Asymmetric encryption enables the use of digital signatures, providing authentication and non-repudiation.
- **Security:** The private key is never transmitted, reducing the risk of it being intercepted.

Disadvantages:

- **Slower:** Asymmetric encryption is slower than symmetric encryption due to the complex mathematical operations involved.
- **Key Size:** To achieve the same level of security, asymmetric encryption often requires much larger key sizes compared to symmetric encryption.

Use Cases:

1. **Secure Communication**: Used in protocols like HTTPS (SSL/TLS) for secure web browsing.
2. **Email Encryption:** Used in systems like PGP (Pretty Good Privacy) for encrypting and signing emails.
3. **Digital Signatures:** Verifying identity and data integrity in software distribution, contracts, and legal documents.

4. **Crypto currency:** Used in block chain technologies like Bit-coin for secure transactions and wallet management.

In summary, asymmetric encryption provides an essential tool for secure communication and digital verification, balancing security and convenience in a world of digital transactions.

Digital Signature

A digital signature is a cryptographic technique used to verify the authenticity and integrity of digital messages or documents. It serves as a way to ensure that a message or document was created by a specific sender (authentication) and has not been altered during transmission (integrity).

Components of a Digital Signature:

1. **Hash Function:**
   - Before signing, the sender generates a hash (or digest) of the message. This is a fixed-length string generated from the original data, which uniquely represents the content of the message.
   - The hash is unique to the message, so any change to the message would produce a different hash.
2. **Private Key:**
   - The sender uses their private key to encrypt the hash of the message. This encrypted hash is the digital signature. The private key is kept secure and never shared.
3. **Public Key:**
   - The receiver can use the sender's public key to decrypt the hash and verify that it matches the hash of the received message. This ensures the message hasn't been altered and confirms that it was signed by the sender.

Process of Digital Signature:

1. **Signing the Message:**
   - **Step 1:** The sender creates a hash of the original message using a cryptographic hash function (e.g., SHA-256).
   - **Step 2:** The sender encrypts the hash with their private key to generate the digital signature.
   - **Step 3:** The sender sends the message along with the digital signature to the recipient.
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2. **Verifying the Signature:**
    - **Step 1:** The recipient receives the signed message and the digital signature.
    - **Step 2:** The recipient uses the sender's public key to decrypt the digital signature, recovering the original hash.
    - **Step 3:** The recipient then computes the hash of the received message and compares it with the decrypted hash. If they match, the message is authentic and hasn't been altered.

Why Digital Signatures Are Important:

1. Authentication: The sender's identity is confirmed because only the sender has access to the private key used for signing.
2. Integrity: If the message is altered in transit, the hashes will not match, signaling that the message has been tampered with.
3. Non-repudiation: The sender cannot deny sending the message because only they possess the private key used to create the signature.

Real-World Applications of Digital Signatures:

- Email Encryption: Services like PGP or S/MIME use digital signatures to ensure email authenticity.
- Software Distribution: Digital signatures are used to sign software or patches to ensure the software hasn't been altered by unauthorized parties.
- Legal Documents: Digital signatures provide a way to sign contracts or agreements electronically, ensuring both the integrity and authenticity of the document.
- Block chain and Crypto currencies: Digital signatures are crucial in securing transactions and verifying ownership in systems like Bit coin.

Digital signatures are essential for secure communication, protecting data from tampering, and ensuring the legitimacy of digital transactions.

Digital Envelope:

A digital envelope is a cryptographic technique that combines both symmetric and asymmetric encryption to securely send data. It leverages the strengths of both types of encryption to ensure confidentiality, integrity, and security in data transmission.

Key Concepts of a Digital Envelope:

1. **Symmetric Encryption:**
   - Symmetric encryption uses the same key for both encryption and decryption. It is fast and efficient for encrypting large amounts of data. However, the main challenge is securely exchanging the encryption key between the sender and recipient.
2. **Asymmetric Encryption:**
   - Asymmetric encryption uses two keys: a public key for encryption and a private key for decryption. It is secure for key exchange but slower compared to symmetric encryption, especially for large datasets.

Structure and Process of a Digital Envelope:

1. **Step 1: Symmetric Key Generation:**
   - The sender generates a random symmetric key (often called a session key). This key will be used for encrypting the actual message or data.
2. **Step 2: Data Encryption:**
   - The sender encrypts the message (or data) using symmetric encryption with the generated session key. This results in the encrypted message.
3. **Step 3: Asymmetric Encryption of the Session Key:**
   - The sender then encrypts the session key using the recipient's public key (asymmetric encryption). This ensures that only the recipient can decrypt the session key using their private key.
4. **Step 4: Sending the Digital Envelope:**
   - The encrypted message (the result of symmetric encryption) and the encrypted session key (the result of asymmetric encryption) are sent together as a digital envelope.
5. **Step 5: Decryption by the Recipient:**
   - Upon receiving the message, the recipient first uses their private key to decrypt the session key (which was encrypted with their public key).
   - Then, the recipient uses the decrypted session key to decrypt the encrypted message (which was encrypted with the session key).

Advantages of Digital Envelopes:

1. **Efficiency:**
   - Symmetric encryption is faster and more efficient for encrypting large amounts of data compared to asymmetric encryption. Using symmetric encryption for the message ensures quick and efficient data transmission.
2. **Security:**
   - By encrypting the session key with the recipient's public key, asymmetric encryption ensures that only the intended recipient can decrypt the session key. This prevents anyone else from intercepting the key.
3. **Combining Strengths:**
   - The digital envelope combines the speed of symmetric encryption for large data and the security of asymmetric encryption for key exchange.

Applications of Digital Envelope

- **Secure Email Communication:**
  - The sender generates a random session key, uses it to encrypt the email content with symmetric encryption, and then encrypts the session key with the recipient's public key. Both the encrypted email and the encrypted session key are sent to the recipient as a digital envelope.
  - Upon receiving the email, the recipient uses their private key to decrypt the session key, and then uses the session key to decrypt the email content.

Diagram of the Digital Envelope Process:

```
+--------------------+          +--------------------+
| Sender Generates   |          | Recipient          |
| a Random Session   |          | Receives Encrypted|
| Key for Symmetric  |          | Message and        |
| Encryption         |          | Encrypted Session |
+--------------------+          | Key                |
          |                     +--------------------+
          |                               |
 +--------------------+          +--------------------+
 | Encrypt Message    |          | Decrypt Session    |
 | with Session Key   |          | Key using Private |
 | (Symmetric)        |          | Key (Asymmetric)   |
 +--------------------+          +--------------------+
          |                               |
 +--------------------+           +--------------------+
 | Encrypt Session    |           | Decrypt Message    |
 | Key with Public    |           | with Decrypted     |
 | Key (Asymmetric)   |           | Session Key        |
 |                    |           |  (Symmetric)       |
 +--------------------+           +--------------------+
          |
 +--------------------+
 | Send Digital       |
 | Envelope (Encrypted|
 | Message + Session  |
 | Key)               |
 +--------------------+
```

A digital envelope is an efficient and secure method for transmitting sensitive information over the internet. It combines symmetric encryption (for fast data encryption) with asymmetric encryption (for secure key exchange), ensuring both data confidentiality and secure key management. This method is widely used in protocols like SSL/TLS, which secure web communications, and other secure messaging systems.

Important Questions (Assignment: II)

Q. 1: Explain symmetric/conventional/classical encryption (Symmetric Cipher Model).

Q.2: Explain in detail asymmetric/public key encryption.

Q.3: Explain structure of Public key encryption.

Q.4: Write notes on following

1) Digital Signature 2) Digital Envelope